

Native pressure-gauge recovery: full hull follow-up

Two independently verified frozen contact systems recover with residuals $4.81\text{e-}9$ and $1.53\text{e-}10$ m/s, using a cold restart and a mechanically neutral pressure redistribution respectively. The native implementation reproduces both results and separately checks the unilateral normal law, circular capacity, maximum dissipation and finite passivity bound.

The full-trajectory study retains **six attempts and zero completed histories**. All six therefore remain unqualified. This is a numerical improvement for two contact systems, not a claim of general random-shape accuracy.

Attempt	Failed velocity residual	Outcome
fast_rotate_shake27_hulls7301/reference_0	9.902e-08 m/s	Rejected
fast_rotate_shake27_hulls7301/reference_1	0.00453103 m/s	Rejected
fast_rotate_shake27_hulls7301/reference_2	1.79796e-05 m/s	Rejected
fast_shake8_hulls42/reference_0	1.06032e-05 m/s	Rejected
fast_shake8_hulls42/reference_1	1.16451e-08 m/s	Rejected
fast_shake8_hulls42/reference_2	3.47786e-05 m/s	Rejected

Execution source: `f6d03878e7d187b18be8d590c6e66f976232e020`. The source and plan were committed before execution. The archives preserve the exact source bytes, authored geometry, checksums and every error. The paired baseline is the gyroscopic-RHS follow-up; geometry, materials, gravity, prescribed translation/rotation, timesteps, travel fractions and physical/trajectory gates are unchanged. Only the numerical contact search gains bounded recovery.

The ordinary iterative solve runs first. Recovery uses the same circular-contact projection equations, a numerical SVD Newton step, mechanically neutral gauge moves to neighboring impulse-boundary faces and a cold Newton restart. It adds no compliance to the physical mobility and substitutes no friction pyramid. Recovery is restricted to 256 rows, at most 64 Newton steps per attempt and a global ceiling of 256 SVD calls, each with at most 64 Jacobi sweeps. Unconverged numerical steps reject. Complete contact residual and finite energy/work gates remain mandatory. Work counters are part of the native output.

The successful frozen systems can still be followed by different, unresolved systems later in a simulation. Those later failures must be diagnosed before promoting an arbitrary-body accuracy preset. The effective contact model remains synthetic and rigid; independent elastic couple/history integration into the many-body engine and experimental material calibration remain open.

Reproduce with `python -m research.run_spatial_recovery` from the frozen source; audit with `python -m research.audit_spatial_recovery`. The captured-system solver and certificates are in `research/coulomb-diagnostics/`.